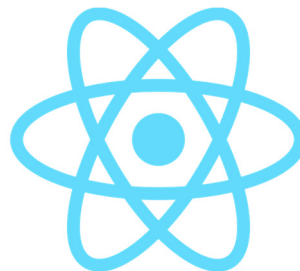


Web Components and Some Web Templates, Frameworks, Platforms, and Libraries



HTML5 ★ BOILERPLATE

A dead simple, responsive boilerplate.



Web components...

- Web components are APIs that allow the creation of new, custom HTML tags (<https://www.webcomponents.org/>)
- Based on four main specifications:
 - *Custom Elements*: the basis for designing and using new types of DOM elements
 - *Shadow DOM*: specifies how to use encapsulated style and markup in web components
 - *ES Modules*: define the inclusion and reuse of JavaScript documents in a standard, modular, and performant way
 - *HTML Template*: specifies how to declare fragments of markup that are not used at page load, but can be instantiated later at runtime



... Web components

Example: defining a `<popup-info>` element

(From https://developer.mozilla.org/en-US/docs/Web/Web_Components/Using_custom_elements)

```
<!DOCTYPE html>
<html>
  <head>
    <title>Trying custom elements</title>
    <script src="custel.js" defer></script>
  </head>
  <body>
    <h1>An example of custom element</h1>
    <form>
      <div>
        <label for="cvc">Enter your CVC</label>
        <popup-info img="img/info.png"
          data-text="Your card validation code (CVC) is an extra security
            feature – it is the last 3 or 4 numbers on the back of your card.">
        </popup-info>
        <input type="text" id="cvc">
      </div>
    </form>
  </body>
</html>
```

See the result (and `custel.js`) at <https://vision.unipv.it/wmt/jsexamples/customel/>

Examples of web templates

Bootstrap, HTML5 Boilerplate, Skeleton

Bootstrap (<https://getbootstrap.com/>)

Open-source toolkit to build responsive, mobile-first projects with HTML, CSS, and JavaScript (originally developed at Twitter)



HTML5 Boilerplate (<https://html5boilerplate.com/>)

Template for building fast, robust, and adaptable web apps or sites

HTML5 ★ BOILERPLATE

Skeleton (<http://getskeleton.com/>)

Simple, responsive template to kickstart any responsive project

A dead simple, responsive boilerplate.

Node.js...

Free, open-source, cross-platform JavaScript runtime environment that lets developers create servers, web apps, command line tools, and scripts



- Built on Chrome's V8 JavaScript engine, it uses an event-driven, non-blocking I/O model that makes it lightweight and efficient for handling many simultaneous connections
- With its extensive “npm-powered” ecosystem offering access to more than two million reusable packages, developers can rapidly build and deploy sophisticated applications

<https://nodejs.org/>

... Node.js ...

Node.js uses JavaScript on the server to enable asynchronous programming

- It must be installed on the computer that will run it
- E.g., file 'first-trial.js':

```
var http = require('http');
http.createServer(function (req, res) {
  res.writeHead(200, {'Content-Type': 'text/html'});
  res.end('Hello World!');
}).listen(8080);
```

To make the computer work as a server:

```
C:\Users\Marco>node first-trial.js
```

... Node.js

Example:

File 'myfirstmodule.js':

```
exports.myDateTime = function () {  
    return Date();  
};
```

http://localhost:8080

File 'mymodule.js':

```
var http = require('http');  
var dt = require('./myfirstmodule');  
  
http.createServer(function (req, res) {  
    res.writeHead(200, {'Content-Type': 'text/html'});  
    res.write("The date and time are currently " + dt.myDateTime());  
    res.end();  
}).listen(8080);
```

The date and time are currently Wed Jan 15 2020 14:54:33 GMT+0100 (Or
standard dell'Europa centrale)

Angular...

Client-side, single-page platform and framework, developed by Google, to create web applications (requires *Node.js* in development)



- It uses the TypeScript language (by Microsoft, a superset of JavaScript), transpiled into JavaScript for client-side execution
- It enables developers to build their components – custom HTML elements that are the building blocks of a user interface. A component is a reusable piece of functionality that can be used in a project and reused in other Angular projects
- It allows to move data from JavaScript to the view and manage user events without writing code
- “Two-way binding”: any interface element change is reflected in the corresponding “model state”, and any model state change is reflected in the user interface state

<https://angular.io/>

... *Angular*...

Example of component

```
import { Component } from '@angular/core';

@Component({
  selector: 'hello-world',
  template: `
    <h2>Hello World</h2>
    <p>This is my first component!</p>
  `
})
export class HelloWorldComponent {
  // The code in this class drives the component's behavior.
}
```

To use the component:

```
<hello-world></hello-world>
```

The component will be rendered as:

```
<hello-world>
  <h2>Hello World</h2>
  <p>This is my first component!</p>
</hello-world>
```

... Angular

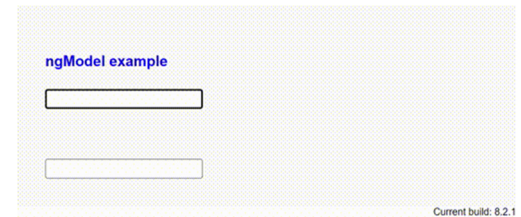
ngModel

- The `ngmodel` directive binds the value of HTML controls to application data: if the user changes the value of an input field, the Angular property will also change its value
- Example:

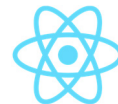
```
In app.module.ts: import { FormsModule } from '@angular/forms';
```

```
In app.component.html:
```

```
<input [(ngModel)]="CopyText">
<strong>{{CopyText}}</strong>
<input type="text" value="{{CopyText}}">
```



React...



Is a JavaScript library for building user interfaces

- Allows to build encapsulated components that manage their own state and can be combined to create complex interfaces
- Example:

```
class HelloMessage extends React.Component {  
  render() {  
    return <div>Hello {this.props.name}</div>;  
  }  
}  
  
root.render(<HelloMessage name="Taylor" />);
```

RESULT

Hello Taylor

<https://reactjs.org/>

... *React*

Example:

```
class Timer extends React.Component {
  constructor(props) {
    super(props);
    this.state = { seconds: 0 };
  }

  tick() {
    this.setState(state => ({
      seconds: state.seconds + 1
    }));
  }

  componentDidMount() {
    this.interval = setInterval(() =>
      this.tick(), 1000);
  }
}
```

```
componentWillUnmount() {
  clearInterval(this.interval);
}

render() {
  return (
    <div>
      Seconds: {this.state.seconds}
    </div>
  );
}
} // end class definition

root.render(<Timer />);
```

RESULT

Seconds: 1399

jQuery

jQuery is a fast, small, and feature-rich JavaScript library



- It simplifies HTML document traversal and manipulation, event handling, animation, and Ajax interactions through an easy-to-use API
- It wraps common tasks that require many lines of JavaScript code into methods that can be called with a single line of code
- Example:

```
var hiddenBox = $("#banner-message");  
$("#button-container button").on("click", function(event) {  
    hiddenBox.show();  
});
```

shows the `#banner-message` element, which is hidden with `display: none` in its CSS, when any button in `#button-container` is clicked

<https://jquery.com/>